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Evaluation of nickel supported vanadium and carbon catalyst for sustainable aviation fuel production from waste cooking oil

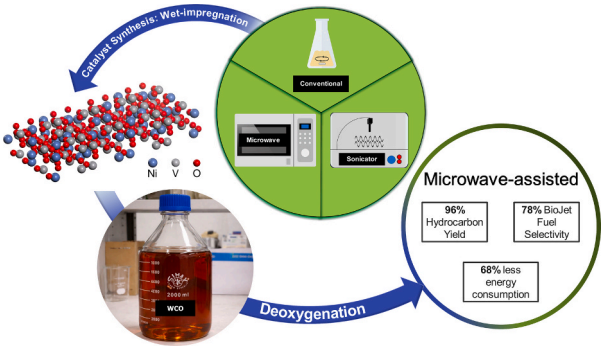
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HIGHLIGHTS

- Enhanced catalyst physicochemical properties achieved via microwave method.
- Microwave-assisted Ni/V_2O_5 catalyst achieved 96% hydrocarbon yield.
- Energy reduced by 68–77% with microwave and sonication synthesis.
- Ni/V_2O_5 shows high jet-fraction selectivity and strong stability.
- TGA and RAMAN confirmed carbon deposition as the cause of Ni/V_2O_5 deactivation.

GRAPHICAL ABSTRACT



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ABSTRACT

Developing sustainable aviation fuel (SAF) is essential for achieving net-zero emissions, yet economic viability remains a major challenge. In this study, the catalytic deoxygenation (DO) of waste cooking oil was investigated using nickel promoted catalyst such as Ni/V_2O_5 and Ni/AC, which were synthesized via wet-impregnation using treatments of microwave irradiation (M), sonication (S) and conventional stirring (C). Physico-chemical properties revealed an M method showed improved surface area, high pore volume, enhanced nickel dispersion, optimized acidity than S and C method, suggesting M method induced the formation of metallic phase through fast, volumetric heating that accelerated reduction, nucleation and crystallization. Similarly, with DO activity, Ni/V_2O_5 M showed comparable or superior performance to C and S method at 95% reduced time, 68% less energy consumption with 96% hydrocarbon liquid yield and 78% jet-fraction selectivity under H_2-free condition at 400 °C for 2 h over 5 wt% loading. Enhanced performance is attributed to the strong metal-support interaction, where Ni-O-V interfacial sites generated rich oxygen vacancies that promote DO while Ni^0 facilitated

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hydrogenation of unsaturated intermediates and hydrocarbons to produce high *n*-paraffins. Kinetic studies confirmed catalyst stability with low deactivation energy (48.3 kJ/mol) and strong Arrhenius correlation (R^2 = 0.94). These findings highlight Ni/V_2O_5 as a promising, energy-efficient catalyst for SAF production from waste feedstocks, offering a scalable route toward sustainable aviation.

1. Introduction

The aviation industry faces an urgent need for sustainable fuels to mitigate climate change. The negative environmental impacts of fossil-based fuels, coupled with their rapid depletion and volatile market prices, have intensified global efforts to develop clean and renewable energy alternatives. The aviation sector faces growing challenges as air travel demand continues to rise, leading to increasing CO_2 emissions that significantly contribute to climate change (Adzahar et al., 2026). Bio-jet fuel (BJF), composed primarily of C_8-C_{16} hydrocarbons, has emerged as a promising sustainable alternative. BJF cuts life-cycle greenhouse gases while maintaining drop-in compatibility, in addition to lower sulfur content, lower tailpipe emissions, high thermal stability, and favourable cold-flow properties (Hebish et al., 2024). Triglyceride oils represent attractive feedstocks for BJF production, as their long-chain fatty acids closely resemble hydrocarbons. However, upgrading and oxygen removal are required, typically achieved through deoxygenation (DO) and hydrodeoxygenation (HDO). HDO relies on substantial amounts of high-pressure hydrogen gas, which raises safety concerns over storage, delivery and operational hazards and undermines sustainability, since clean H_2 production is far from commercialization and thus approximately 96% of global hydrogen is currently derived from non-renewable sources such as naphtha natural gas reforming. In contrast, DO does not require external H_2 input; instead, oxygen is removed through decarboxylation (CO_2 release) or decarbonylation (CO and H_2O release) collectively referred to as $DeCo_x$ pathways, offering a more sustainable H_2-free pathway for BJF production. Note, CO and CO_2 gases by-products of DO reaction can be captured and not released into the environment. On an industrial or pilot scale with a complete system, these gases will be recycled back and used as feedstocks to produce other valuable chemicals and intermediates such as methane, urea, syngas etc (James et al., 2023; Zhang et al., 2020). Several studies have reported on the promising potential of DO reactions. For instance, Why et al., 2022 reported a 96% liquid yield and 73% jet fuel selectivity for the DO of palm kernel oil under Pd/C. In another study, Asikin-Mijan et al., 2022 observed 77% hydrocarbon yield and 64% jet fuel selectivity for the DO of tung oil under CoW/SiO_2-Al_2O_3. Furthermore, the DO of soybean oil under $NbOPO_4$ was found to yield 91% liquid product and 58% jet-range selectivity (Scaldaferri and Pasa, 2019). Despite these advances, catalyst efficiency and energy consumption remain critical challenges.

Waste Cooking Oil (WCO) is a major byproduct of food preparation, and improper disposal poses serious environmental hazards. Globally, around 16.5 million tonnes of WCO are generated annually, creating challenges for environmentally safe disposal (Sharma et al., 2021). Palm-based WCO, comprising mainly of palmitic acid (C16:0) and oleic acid (C18:1), presents an attractive feedstock to produce BJF. In fact, WCO can produce highly competitive fuel and potentially most cost-effective, along with avoiding disposal hazards and avoiding direct competition with edible oil feedstocks in the food supply chain (Detsios et al., 2023).

Although noble metal catalysts exhibit excellent DO performance, their prohibitive cost and scarcity limit large-scale application (Hebish et al., 2024). Similarly, sulfided catalysts, despite their effectiveness, face challenges such as sulfur leaching and contamination of the final product, making them less desirable as catalysts for biofuel synthesis (Gamal et al., 2020). Consequently, this has driven the search for affordable, sulfur-free alternatives. In this regard, transition metals and their oxides have emerged as promising alternatives. Among them, Ni-based catalysts offer a sustainable and industrial relevant alternative in upgrading of triglyceride oils for biofuel production owing to their relatively low cost and scalability and demonstrated strong hydrogenation-DO capability and high catalytic activity (Aliana-Nasharuddin et al., 2020; Hafriz et al., 2022; Hartono et al., 2023; Khalit et al., 2021). As reported by Morgan et al., Ni showed higher DO and cracking activity than Pd/Pt on similar supports, yielding more C_8-C_{17} hydrocarbons (Morgan et al., 2010). Similar behaviour was observed by Jeon et al. during the DO of oleic acid over a Ni-$CeZrO_2$ catalyst, which attained ~ 98% conversion C_9-C_{18} hydrocarbon fraction of 34% and an oxygen removal efficiency of ~ 96% (Jeon et al., 2019). Furthermore, in a comparative study conducted by Hafriz and colleagues, transition metals Ni, Zn, Cu, Co and Fe were evaluated as catalyst promoters. Among these catalysts, findings reveal the Ni-based catalyst system exhibited the highest conversion activity (67%), high portion of saturated hydrocarbons (80%), and highest selectivity towards green diesel (55%) and BJF (39%) (Hafriz et al., 2020). Collectively, these studies clearly demonstrate that Ni-promoted catalysts enhance DO activity. Among Ni-based catalyst, Ni supported on activated carbon (AC) shows great potential. Previously, Wu et al. performed DO on stearic acid over Ni/AC achieving an almost complete conversion with heptadecane (C_{17}) selectivity of ~ 80% (Wu et al., 2016). The research group of (Khalit et al., 2021) tested Ni catalyst over various supports of AC, zeolite, and reduced graphene oxide for the DO of WCO to produce green diesel. In their work, Ni/AC was reported the most active catalyst by giving 90% hydrocarbon yield and 89% selectivity towards *n*-(C_{15} + C_{17}) owing to its good physicochemical properties of porosity, high surface area and high number of acid-base sites. It is noteworthy that, vanadium pentoxide (V_2O_5) is widely recognized as a cost-effective, thermally stable catalytic material with excellent redox and oxygen transfer properties that has great value in the industry (Marberger et al., 2019). Despite this, as far as we know the utilization of V_2O_5 support remains unexplored in DO reaction.

Recent studies show promising results in microwave irradiation (M) and sonication (S)-assisted synthesis techniques of active catalysts for biofuel upgrading, whereby these treatments resulted in enhanced physicochemical properties and catalytic performance (Luo et al., 2022). M method induces volumetric heating that leads to uniform temperature distribution, accelerated nucleation and improved metal dispersion on support. Furthermore, atoms movement occurs where angles of atoms will shift and open up the active sites. It is also reported that catalyst exhibited more porosity and higher surface area following microwave exposure (Burakova et al., 2018). For instance, Xue et al. reported the development of Ni_2P/SiO_2 catalysts using M method and the result shows enhanced dispersion of Ni_2P particles on support and catalytic performance in the HDO of methyl palmitate (Xue et al., 2015). Similarly, ultrasonic approaches exploit acoustic cavitation to induce localised hotspots and micro-mixing resulting in enhanced catalyst surface properties and metal dispersion and thus improve catalytic activity (Li et al., 2021). In a study conducted by (Ameen et al., 2019) Ni-Mo/γ-Al_2O_3 catalyst synthesis by ultrasound achieved significantly higher hydrocarbon yields (80.87%) during the HDO of rubber seed oil compared to conventional prepared catalyst (63.3%).

To the best of the authors' knowledge, no systematic study has directly compared M and S treatments of the same catalyst formulation for triglyceride DO with focus on jet-fuel range hydrocarbons. Furthermore, although Ni-based catalysts have been extensively studied and V_2O_5 exhibits excellent potential as a catalyst support, the Ni/V_2O_5 bimetallic catalyst remains largely unexplored, despite its potential synergistic effects that could significantly enhance catalytic activity.

Due to this reason, hence the present study, we developed a series of inexpensive Ni/V_2O_5 and compare with performance of established effective Ni/AC catalysts, using different synthesis treatments (conventional, microwave and sonication), for the effective DO of WCO into biojet fuel product. To evaluate the effect of the treatments, the catalysts properties were studied through various characterizations, and the catalytic performance was assessed by analysing their respective liquid product with regards to BJF selectivity. Additionally, kinetic deactivation study was also carried out to provide insights into catalyst deactivation to assess the catalyst further.

2. Materials and methods

2.1. Materials

Commercial Vanadium Oxide (V_2O_5) with purity > 99.6% was purchased from Sigma-Aldrich, USA. Nickel (II) Nitrate Hexahydrate (Ni$(NO_3)_2{\cdot}6H_2O$) with 99% purity, Activated Carbon (AC) powder was supplied by Chemiz (M) Sdn. Bhd, Malaysia and Sulfuric Acid (98% purity) was procured from J.T. Baker, USA and *n*-Hexane GC grade procured from Supelco, Germany. Waste cooking oil (WCO) derived from palm oil feedstocks was obtained from a local street vendor in Kuala Lumpur, Malaysia. The oil obtained was left to rest for 48 h before carefully decanting to leave behind large impurity solids. Whatman grade 1 (11 μm) filter paper was used to filter finer impurities. Preheating at 110 ℃ for 2 h was necessary to remove water content. The analysis of fatty acid composition of WCO is exhibited in Table 1. The acid value and free fatty acid (FFA) were quantified, and the results were consistent with the typical range reported in the literature (Plata et al., 2022).

Catalysts Synthesis and Treatments.

Ni/V_2O_5 was prepared using the wet-impregnation method where a 1:6.5 solid-to-liquid ratio was employed based on the support water solubility of 515 g/l according to Sigma Aldrich (USA). A nickel loading of 20 wt% was selected based on prior studies, whereby this loading was associated with high density of accessible Ni sites and high catalytic rate without excessive sintering or pore blockage (Jeon et al., 2022). Therefore, 9.92 g of Ni$(NO_3)_2{\cdot}6H_2O$ was dissolved in 15.53 ml of deionised water to produce Ni salt (20 wt%) which was added dropwise into 8 g of V_2O_5 and mixed continuously at room temperature for approximately 12 h impregnation time, this was denoted as Ni/V_2O_5 Conventional (C). In another instance the Ni salt (20 wt%) was incorporated into the V_2O_5 and exposed to 80 W microwave irradiation at 210 rpm stirring speed for 30 min with 5 s on and 25 s off cycles using a household microwave oven (Sharp, Japan) and left to rest for 1 h, this was marked as Ni/V_2O_5 Microwave (M). Another case, the impregnated sample was subjected to sonication performed at 60% amplitude (500 W, 20 kHz) for 30 min, utilizing a pulsed mode with 5 s on and 25 s off cycles and left to rest for 1 h, this was represented as Ni/V_2O_5 Sonicator (S). Similar procedures were carried out for Ni/AC C, M and S, respectively. However, prior to use, the commercial activated carbon underwent chemical activation by reflux with sulfuric acid at 150 °C under a nitrogen atmosphere for 12 h. The sample was subsequently washed with warm distilled water (>85 ℃) until a neutral pH (pH = 7) was achieved, ensuring the removal of residual sulfate ions and other unwanted impurities. Finally, it was subjected to drying in an oven at 100 ℃ for 12 h. All impregnated samples were dried at 100 ℃ for 24 h. Additionally, the catalysts underwent a necessary pre-treatment step of calcination where V_2O_5 was calcined at 600 ℃ for 4 h in air, while Ni/AC and Ni/V_2O_5 were calcined at 600 ℃ for 4 h in a N_2 inert environment, to boost their activity for the impending DO process. For comparison, the estimated energy consumption of each synthesis route is summarized in Table 2. It is worth noting that M and S methods required approximately 95% less time compared to conventional stirring, while also consuming significantly lower energy, with reductions of about 67–77%.

2.2. Catalysts characterization

The specific surface area, pore volume, and pore size distribution of the samples were determined using nitrogen adsorption–desorption isotherms measured at liquid nitrogen temperature (−196 °C). The analysis was performed using a Micromeritics TriStar II PLUS. Prior to analysis, the samples were degassed under vacuum at 150 °C for 4 h, to remove any impurity gas or vapours which may have adsorbed onto the surface from the ambient air. The surface area was calculated using the Brunauer–Emmett–Teller (BET) method, while the pore size distribution was obtained using the Barrett–Joyner–Halenda (BJH) method. The crystalline phases and structural properties of the catalysts were analysed using powder X-ray diffraction (XRD). The measurements were carried out on a Bruker D8 Advance equipped with Cu Kα radiation (λ = 1.54060 Å), operated at 40.0 kV and 40.0 mA. Scans were conducted over a 2θ range of 5°- 80° at a scanning rate of 0.0384°/min. The resulting diffractograms were compared with standard reference patterns from the International Centre for Diffraction Data (ICDD) to identify crystalline phases, and crystallite size was calculated using Debye-Scherrer equation. Thermal stability of the samples was examined using thermogravimetric analysis (TGA). The analysis was performed using a Mettler Toledo TGA/SDTA 851e. Approximately 10 mg of sample was placed in a ceramic crucible and heated 25 °C to 900 °C at a heating rate of 30 °C/min under a N_2 flow of 40 mL/min. The acidity of the catalyst was evaluated using temperature programmed desorption with ammonia (TPD-NH_3) analysis on a Micromeritics AutoChem II Plus. The sample was pretreated in N_2 at 300 °C for 30 min, then cooled to 50 °C and exposed to a 5% NH_3/He gas mixture for 30 min. After flushing with He for another 30 min, the temperature was ramped up to 900 °C, and the desorbed ammonia was detected by a thermal conductivity detector (TCD). The catalyst reducibility was assessed using Temperature programmed reduction (H_2-TPR) analysis on a Micromeritics AutoChem II Plus. The sample was first pretreated in N_2 flow of 20 cm^3/min at 150℃ for 1 hr, followed by cooling to ambient temperature. The catalyst was then subjected to a 10% H_2/N_2 gas mixture, and the temperature was increased to 900℃ at a rate of 15 ℃/min while H_2

Table 1
Fatty acid composition of WCO*.

Properties	WCO
FFA Content (%)	0.55 ± 0.04
Acid Value (mg KOH/g)	1.1033 ± 0.07
Fatty Acid Composition (%)	
Myristic acid (C14)	0.91 ± 0.37
Palmitic acid (C16:0)	34.42 ± 0.21
Palmitoleic acid (C16:1)	0.60 ± 0.20
Margaric acid (C17:0)	0.10 ± 0.07
Oleic Acid (C18:1)	59.36 ± 0.36
Others	4.61 ± 0.44

* Values are reported as mean ± standard deviation (n = 3).

Table 2
Energy consumption of different synthesis methods*.

Synthesis Method	Equipment	Power, P (W)	Time, t (h)	Total energy Consumption (kWh) = P*t
Conventional	Magnetic Stirrer	15.0 ± 0.8	12.00 ± 0.00	0.180 ± 0.010
Microwave-assisted	Microwave Oven	80.0 ± 4.0	0.50 ± 0.00	0.058 ± 0.003
	Overhead Stirrer	35.0 ± 1.8	0.50 ± 0.00	
Sonication-assisted	Sonicator	500.0 ± 25.0	0.08 ± 0.000	0.042 ± 0.002

* Values are reported as nominal ± uncertainty.

consumption was monitored by a TCD detector. The surface morphology and elemental composition of the samples were analysed using scanning electron microscopy (SEM) coupled with energy dispersive X-ray spectroscopy (EDX). The measurements were performed on a SEM Phenom ProX, Thermo Fisher operated at an accelerating voltage of 15 kV. Samples were coated with a thin layer of gold. Elemental mapping was conducted using EDX to determine the elemental distribution and composition. Moreover, for the kinetic deactivation study, the deposited coke on the spent catalysts surface was characterized using TGA (Toledo TGA/SDTA 851e), heating from 25 ℃ to 1000 ℃ with a ramp of 30 ℃/min under 40 mL/min of O_2 and RAMAN spectroscopy (Renishaw inVia Microscope) with a laser excitation (514 nm).

2.3. Catalytic DO reaction

The experimental setup for the deoxygenation (DO) process, as illustrated in Fig. S1, comprised a 500 mL triple-necked semi-batch reactor coupled with a condensing system. Initially, the non-pressurized reactor was purged with N_2 to establish an inert environment by eliminating all O_2. Subsequently, 15 g of WCO and 5 wt% catalyst were introduced into the reactor. The reaction mixture was then subjected to controlled heating and continuous stirring at 400 ℃ and 300 rpm using a heating mantle and stirrer equipped with a PID temperature controller, while maintaining a steady N_2 gas flow of 20 mL/min for 2 h in atmospheric pressure. Throughout the reaction, vaporized products were condensed via a cooling chamber maintained at 20 °C, with the resulting liquid deoxygenated product collected in a small vessel, weighed and subsequently analysed by gas chromatography with a flame ionization detector (GC-FID), gas chromatography-mass spectrometry (GC–MS) and Fourier transform infrared spectrometry (FTIR). The gaseous products were collected after 1 h reaction time and further analysed by gas chromatography with thermal conductivity detector (GC-TCD). To ensure statistical reliability, the DO process was replicated thrice, and the average result was calculated.

2.4. Liquid product analysis

The hydrocarbon yield was evaluated by GC-FID using an AGILENT 7890A coupled with a HP-5 capillary column (0.25 μm x 0.32 mm x 30 m) and utilising alkane and alkene (C_8-C_{20}) standard acquired from Sigma Aldrich (USA). Prior to analysis, the deoxygenated liquid product was diluted with GC-grade *n*-hexane, and ~ 2 μL of sample was injected into the GC system equipped with an FID detector. Helium was used as the carrier gas. The injector and detector temperatures were set at 250 ℃ and 300 ℃, respectively. The oven temperature was initially held at 40 ℃ for 6 min, then ramped to 270 ℃ at a rate of 7 ℃/min. Each sample was analysed in triplicate to ensure reproducibility The concentration of various hydrocarbons (jet-range hydrocarbons: *n*-paraffin and olefin, naphthene, aromatic and heavy hydrocarbon (C > 16)) and byproducts or others (e.g. ketone, alcohol, and carboxylic acid) were identified with GC–MS, using SHIMADZU GC-TQ8030 equipped with a SH-5MS capillary column (thickness: 0.25 μm x length: 30 m x diameter: 0.25 mm), with analysis carried out under similar operating conditions to GC-FID. Compounds were identified from the GC–MS spectra using the National Institute of Standards and Technology (NIST) library based on a probability match of 95% or higher. Fourier transform infrared (FTIR) spectroscopy was performed using a Shimadzu IRAffinity-1S spectrophotometer. Spectra were recorded in the range of 650–4000 cm^{-1} to identify the chemical functional groups present in the liquid product. Gas products collected were analysed by the Agilent USA model GC6890 N equipped with a TCD via an offline sampling method to determine the composition and concentration of the gases produced during reaction. The mass yield, hydrocarbon yield and jet-selectivity were calculated using Eqs. (1) to (3), respectively.

$$Mass\ Yield\,(\%) = \frac{mass\ of\ liquid\ product}{mass\ of\ WCO} \times 100 \tag{1}$$

$$Hydrocarbon\ Yield\,(\%) = \frac{\Sigma\ area\ of\ alkane\,(C_8 - C_{20}) + \Sigma\ area\ of\ alkene\,(C_8 - C_{20})}{\Sigma\ area\ of\ hydrocarbon\ product} \tag{2}$$

$$BJF\ Selectivty\,(\%) = \frac{\Sigma area\ of\ alkane\,(C_8 - C_{16}) + \Sigma area\ of\ alkene\,(C_8 - C_{16})}{\Sigma area\ of\ hydrocarbon product} \tag{3}$$

2.5. Kinetic deactivation study

The catalyst stability of Ni/V_2O_5 M was investigated at different temperatures (380 ℃, 400 ℃ and 420 ℃) for the DO of WCO and the relative rate was plotted against time-on-stream as shown in Fig. 3a-c. The ratio of hydrocarbon yield (C_8-C_{20}) at time t, CHC_t to the initial hydrocarbon yield (C_8-C_{20}) at time = 0, CHC_0 is referred to as the relative catalytic activity, a_t. Based on previous studies, the Levenspiel power-law deactivation model is a widely used approach for systems with relatively fast catalyst deactivation such as those used for biofuel conversion (Shakor and Al-Shafei, 2023). In this modelling approach, several assumptions were made; single dominant deactivation mechanism of coke formation resulting in blocking or fouling active sites, no catalyst regeneration during reaction, the relative hydrocarbon yield is directly proportional to catalyst activity and at t = 0, catalyst is fully activated (i.e.a_0 = 1). Given the assumptions, the model can be expressed as follows Eq. (4).

$$a_t = [1 + (m-1)k_d t]^{-\frac{1}{m-1}} \tag{4}$$

Where a_t is the relative catalytic activity ($a_t = \frac{CHC_t}{CHC_0}$), k_d is the deactivation rate constant, m is the deactivation order and t is the time-on-stream (min). The model activity profile was fitted by obtaining the parameters (k_d and m) using the GRG non-linear least-squares regression via Excel SOLVER, which minimized the sum of squared errors (SSE) between experimental and model-predicted activity. The experimental results show a progressive decline in relative catalytic activity across all temperatures, which can be attributed to coke generation and its accumulation on the catalyst surface, leading to pore blockage and active-site deactivation. The results reveal that the 2nd order power-law model (m = 2) produced the best fitting as shown. The goodness of fit was assessed by the coefficient of determination R^2 yielding acceptable values (0.81–0.98). Furthermore, Arrhenius plot was drawn to determine the activation energy of deactivation, E_d, which reflects the temperature dependence of the deactivation rate (Shakor and Al-Shafei, 2023). We assume the deactivation rate constant follows the Arrhenius form.

$$k_d(T) = Ae^{-E_d/(RT)} \tag{5}$$

$$lnk_d = lnA - \frac{E_d}{R}\cdot\frac{1}{T} \tag{6}$$

where k_d is the deactivation rate constant, A is the frequency factor, E_d is the activation energy of deactivation, R is the universal gas constant and T is temperature in kelvin. The activation energy of deactivation (E_d) was determined using the Arrhenius equation by plotting $\ln k_d$ against 1/T.

3. Results and discussion

3.1. Characterization of synthesised catalysts

XRD patterns (Fig. S2a-b) confirm orthorhombic V_2O_5 (JCPDS card no 01–075-9978), and nickel vanadate phases such as $Ni_3(VO_4)$, Ni(V_2O_6) and $Ni_2(V_2O_7)$ in all Ni/V_2O_5 catalysts. The absence of distinct metallic Ni or NiO indicates the formation of strong Nickel Vanadium Oxide interactions during synthesis and calcination, producing mixed oxide phases instead of separate Ni or NiO domains, this is in line with previous reports where incorporation of Ni into oxide supports suppresses Ni and NiO reflections in XRD (Adzahar et al., 2026). In the case of Ni/AC catalysts result reveals sharp reflections indexed to metallic Ni (FCC, JCPDS card no. 00–004-0850) implying a greater proportion of crystalline Ni^0 sites along with the broad band of graphitic carbon at about $2\theta = 26^\circ$. Trace NiO (JCPDS card no. 01–078-0643) is only detected in the C sample, whereas both M and S exhibit exclusively metallic Ni, implying that M and S treatments favour full reduction of Ni species during synthesis likely due to the rapid heating and accelerated nucleation of M and S-assistance. This observation was previously confirmed by Lim et al., 2021 from using H_2-TPR where they reported that sonicated catalysts display more reducible NiO, indicating a higher fraction of Ni can be reduced to metallic Ni (Ni^0). A similar outcome for microwaved catalysts was revealed by (Aimdate et al., 2021). Furthermore, the crystallite size was determined at the most intense peak at $2\theta = 44.5^\circ$ (Ni) and $2\theta = 20.4^\circ$ (V_2O_5) for Ni/AC and Ni/V_2O_5 catalysts, respectively (Table 3). In terms of Ni/AC catalysts, crystallite size followed the decreasing trend S > C > M, suggesting that M method improved Ni dispersion and availability of active sites. Surprisingly, the crystallite size of Ni on Ni/AC obtained from S method was the largest (24 nm) among the rest AC supported catalyst. Former literature reports that for carbon-based supports dispersion quality depends on sonication energy density, too low energy leaves large agglomerates and too high causes re-agglomeration, thereby increasing crystallite size (Rennhofer and Zanghellini, 2021). For Ni/V_2O_5 catalyst crystallite size decreased in the order C > S > M, revealing greater dispersion by M system. Note, C method for both catalyst series showed to favour the formation of large crystallite size and reduced dispersion. Remarkably, the notable decrease in crystallinity for Ni/V_2O_5 M suggests greater dispersion resulting in finer crystallites and greater lattice imperfections, which can be attributed to the rapid disorderly atom arrangement due to microwave volumetric heating, therefore promoting the generation of more interfacial Ni-O-V sites which are reported to exhibit Ni^{2+}/Ni^{3+} and V^{5+}/V^{4+} redox couples with lattice oxygen participation and likely originated from $Ni_3(VO_4)$, Ni(V_2O_6) and $Ni_2(V_2O_7)$ phases (Li et al., 2020). Such redox-active sites are known to promote C-—O bond cleavage and oxygen removal facilitating $DeCO_x$ pathways (Song et al., 2023).

BET analysis (Table 3, Fig. S2c-f) shows synthesis methods strongly influence surface area and porosity. As anticipated, the Ni/AC catalyst series exhibited the highest surface area, owing to the inherent porous structure of carbon. The BET surface area revealed was ~ 0.71—–2.56 m^2/g and 918.95—–997.87 m^2/g for Ni/V_2O_5 and Ni/AC series, respectively. As expected, V_2O_5 crystallizes in dense layered structures which limits the formation of pores and results in low surface area (often < 5 m^2/g) (Marberger et al., 2019). Interestingly, Ni/V_2O_5 M demonstrated the ability to retain a surface area close to that of the V_2O_5 support, suggesting microwave improved dispersion compared to C and S methods. Conversely, S methods resulted in detrimental surface area and porosity, likely due to agglomeration caused by ultrasonic wave cavitation. The N_2 adsorption–desorption test shown in Fig. S2(c) carried out shows Ni/V_2O_5 series represented a hysteresis type III H3 characteristic of slit-shaped pores which was corroborated by SEM images. In contrast, Ni/AC series displayed a hysteresis type IV H4 (Fig. S2 (d)) linked to microporous materials with narrow slit-like pores, as confirmed in SEM images (see Fig. S5c-d). Indeed, this finding in agreement with XRD findings, whereby the M method capable of improving the metal dispersion, preserving surface area and enhancing active site accessibility. As been noted before (Horváth et al., 2024), doping metals on a support typically reduces surface area and porosity of

Table 3
Textural properties and acidity of fresh synthesized catalysts*.

Catalyst	XRD	BET Textural Properties			TPD-NH_3 Acidity		
	Crystallite size[a] (nm)	Surface area[b] (m^2/g)	Total pore volume[c] (cm^3/g)	Average pore size[c] (nm)	Temperature[d] (°C)	Amount of NH_3 adsorbed[d] (μmol/g)	Total number of acidic sites[d] (μmol/g)
V_2O_5	62.3[e] ± 2.7	2.62 ± 0.02	0.0062 ± 0.0003	14.95 ± 0.73	413.0 ± 14.1 676.1 ± 8.1 889.6 ± 11.6	88.4 ± 7.0 31.8 ± 3.2 90.8 ± 8.5	211.0 ± 12.5
Ni/V_2O_5 C	77.4[e] ± 4.2	1.90 ± 0.01	0.0082 ± 0.0000	34.18 ± 0.25	398.2 ± 6.0 659.8 ± 5.5 884.4 ± 8.7	77.8 ± 9.9 9.1 ± 3.3 23.9 ± 4.1	110.8 ± 11.3
Ni/V_2O_5 M	60.8[e] ± 2.4	2.56 ± 0.02	0.0120 ± 0.0001	29.22 ± 0.33	405.1 ± 2.8 768.6 ± 4.4	151.6 ± 4.8 ± 365.4 ± 2.8	517.2 ± 9.5
Ni/V_2O_5 S	63.9[e] ± 3.5	0.71 ± 0.01	0.0031 ± 0.0001	27.33 ± 0.96	418.3 ± 6.4 613.7 ± 11.3 892.2 ± 9.4	22.0 ± 7.3 27.7 ± 4.9 80.7 ± 6.6	130.4 ± 10.0
Ni/AC C	21.2[f] ± 4.0	961.99 ± 0.03	0.2730 ± 0.0004	4.37 ± 0.01	688.0 ± 8.5 874.8 ± 6.2	1612.3 ± 3.2 255.3 ± 2.9	1867.6 ± 4.5
Ni/AC M	19.1[f] ± 2.9	997.87 ± 0.01	0.1070 ± 0.0002	3.07 ± 0.01	682.4 ± 7.7 899.3 ± 5.0	976.4 ± 8.4 137.4 ± 5.1	1113.8 ± 7.9
Ni/AC S	23.5[f] ± 3.6	918.95 ± 0.02	0.3052 ± 0.0001	5.05 ± 0.00	661.2 ± 13.1 901.0 ± 6.7	1113.7 ± 3.3 375.0 ± 5.4	1488.7 ± 6.1

* Values are reported as mean ± standard deviation (n = 3).
[a] Obtained from Debye Scherrer equation
[b] Obtained from Brunauer-Emmett-Teller (BET) theory analysis.
[c] Obtained from Barrett–Joyner–Halenda (BJH) analysis.
[d] Obtained from Temperature programmed desorption of ammonia.
[e] Crystallite size calculated at $2\theta = 20.4^\circ$.
[f] Crystallite size calculated at $2\theta = 44.5^\circ$.

the catalyst as metal particles fill in the voids of the support. However, Ni/V_2O_5 (27.33—34.18 nm) series showed substantially larger pore sizes than its parent support (V_2O_5 (14.95 nm)). According to literature, this can be attributed to metal dopant facilitating distortion and defect formation of the V-O lattice which promotes interlayer expansion leading to a more open pore structure (Liu et al., 2018). Indeed, larger pore size is needed for eliminating mass transfer limitation with power of reactant movement is (>7x), thereby, allow the reactants to access more active sites, subsequently, increasing overall catalyst performance (S. Wang et al., 2023). Overall, this highlights that larger pore size can be more critical than high surface area properties, in terms of DO activity, by improved accessibility and mass transport for bulky WCO reactants.

The H_2-TPR profiles of V_2O_5 and Ni/V_2O_5 catalyst series, shown in Fig. S3(a), reveal that the incorporation of Ni significantly enhances the reducibility of V_2O_5. While pure V_2O_5 exhibits high-temperature reduction peaks (~580-830℃), Ni/V_2O_5 catalysts display prominent low-temperature peaks (~490-580℃) indicating the presence of more easily reducible oxygen species. This shift to lower temperature suggests that Ni addition facilitates the formation of oxygen vacancies (O_v), particularly at the Ni-O-V interface sites. The higher peak area of Ni/V_2O_5 M and therefore the higher total H_2 consumed implies greater concentration of reactive O_v, providing more active site for enhanced DO. The rapid microwave heating promotes accelerated nucleation leading to lattice imperfections that generate Ni-O-V interface sites with rich O_v.

The H_2-TPR profiles of Ni/AC catalyst series is depicted in Fig. S3(b). Indeed, all catalysts exhibited a single peak with varying intensity and broadness. Ni/AC C exhibited a broad and intense reduction peak centred at ~ 285℃ which indicates the presence of relatively high amount of reducible NiO species alongside Ni. This observation corroborates the XRD results, where NiO species in C method were present above the detection limit and therefore identified. Interestingly, the M and S methods display substantially lower hydrogen consumption with reduction peaks centred at 220℃ and 250℃, respectively. The weaker peak intensity and lower temperature are attributed to the presence of minimal NiO species that have formed a thin layer on the surface due to partial oxidation of metallic Ni upon exposure to air. Thus, the nickel phase is predominantly in the metallic state. This result was also in accordance with XRD findings, where trace NiO was below detection limit and therefore not identified.

Catalyst acidity is crucial for facilitating the cleavage of both C-—O and C–—C bonds during the DO reaction (Asikin-Mijan et al., 2017). Therefore, acidity characteristics of the synthesized catalyst were evaluated by TPD-NH_3, and the profiles for the Ni/V_2O_5 and Ni/AC catalyst series are presented in Fig. S3(c) and (d), respectively, and total acid sites are summarized in Table 3. From the outset, the acidity profiles show significant variations influenced by support type and treatment method. All the catalysts displayed three principal desorption regions corresponding to weak acid sites (< 250 ℃), medium acid sites (250 ℃ – 500 ℃) and strong acid sites (> 500 ℃). The total acid sites of the catalysts are in the increasing order, Ni/V_2O_5 C < Ni/V_2O_5 S < V_2O_5 < Ni/V_2O_5 M < Ni/AC M < Ni/AC S < Ni/AC C. Indeed, the acid rich species of activated carbon is attributed to the presence of surface oxygen functional groups, including lactonic, carboxylic, and phenolic groups (Khalit et al., 2021). These organic groups are ascribed to the acidification of AC via concentrated acid, this observation is in line with the findings of Asikin-Mijan et al. and Gamal et al. (Asikin-Mijan et al., 2017; Gamal et al., 2020). It is important to note that an excessive density of acidic sites can promote undesirable side reactions, leading to accelerated coke formation and consequently severe catalyst deactivation (Aliana-Nasharuddin et al., 2020; Khalit et al., 2021). Thus, higher acidity in Ni/AC catalysts may enhance DO activity but risks coke formation. From the TPD-NH_3 graph (Fig. S4), original V_2O_5 is acidic which can be attributed to NH_3 desorbed from Lewis sites (T_{max} ≈ 150 ℃), weak Brønsted sites (T_{max} ≈ 410 ℃) and strong Brønsted sites (T_{max} ≈ 680 ℃) (Mulyatun et al., 2024). The Brønsted sites are mainly associated with V^{4+}–OH groups, while Lewis acidity originates from exposed V^{5+} centres capable of accepting electron pairs (Zhao et al., 2018). It is observed that the integration of Ni into V_2O_5 and the treatment process influenced the degree of Ni dispersion (see XRD) and the accessibility of acid sites, hence altered the final acidity profile. For instance, Ni/V_2O_5 C showed the lowest acidity (111 μmol/g) likely due to agglomerated Ni species partially blocking acidic sites as a result of using bare stirring. In the contrary, Ni/V_2O_5 M displays the highest acidity (517 μmol/g) owing to the rapid nucleation and finer Ni dispersion that preserves surface VOH groups while likely exposing additional Lewis sites. Ni/V_2O_5 S yielded intermediate acidity, consistent with moderate improvement in metal distribution. Ni/AC catalysts exhibited a different acidity pattern, possessing a substantial amount of strong superacid sites (T > 500 ℃). Among these, Ni/AC C showed the highest acidity (1867 μmol/g), followed by S and M, indicating that conventional stirring, for Ni/AC catalyst series, enhanced Ni anchoring and density of accessible Brønsted and Lewis superacid sites.

Fig. S3e-f shows the thermal stability of the Ni/V_2O_5 and Ni/AC catalysts respectively. For Ni supported on V_2O_5 catalysts, thermal degradation was minimal with a mere total weight loss of 1%. Regarding Ni/V_2O_5, the mass profiles (Fig. S3(e)) of microwave- and conventionally prepared samples were nearly identical indicating that both methods produced stable oxide surfaces. As for Ni/AC, the figure shows clear synthesis effect on thermal stability of the AC-supported catalyst. The mass profiles of these catalysts can be categorized into three distinct regions; region I (low temperature) which corresponds to weight loss from trapped moisture and volatiles removal, region II (moderate temperatures up to ~ 700 ℃) weight loss in this region is often attributed to surface groups decomposition and region III where high temperature loss is linked to pyrolysis of more stable carbonaceous species and structural changes. Ni/AC M exhibits noticeably greater mass loss than Ni/AC C, with the later retaining 10% more weight. This is in agreement with reports that microwave treatment introduces additional oxygenated surface groups and disrupts the carbon matrix structure (Aimdate et al., 2021).

The SEM imaging and EDX elemental mapping are illustrated in Fig. S5 and tabulated in Table 4. For Ni/V_2O_5 catalysts, Ni/V_2O_5 C exhibited large, block-like crystalline structures which have been associated in prior studies with high crystallinity and limited metal dispersion (Yan et al., 2024), whereas Ni/V_2O_5 M shows smaller, layered plate-like features likely caused by microwave induced fragmentation and improved dispersion. For sonicated Ni/V_2O_5, more irregular crystalline structures and agglomerated particles were observed. EDX elemental mapping confirms the presence of Ni, V and O in all samples but highlights difference in metal distribution. For instance, Ni/V_2O_5 C exhibits Ni concentrated in localized patches or hot spots, implying partial agglomeration while Ni/V_2O_5 M demonstrates a notably uniform Ni distribution where the Ni species (~19%) was found to be comparable to the theoretical value (20%). Ni/V_2O_5 S shows moderate dispersion with some residual clustering. For Ni/AC catalyst, all samples retain the characteristic porous texture of activated carbon, however Ni/AC C

Table 4
EDX elemental composition of catalysts*.

Catalyst	Elemental composition (%)			
	Experimental			
	Ni	V	O	C
Ni/V_2O_5 C	15.70 ± 1.54	45.77 ± 2.01	38.54 ± 1.87	–
Ni/V_2O_5 M	18.96 ± 1.94	45.62 ± 0.33	35.42 ± 1.62	–
Ni/V_2O_5 S	24.63 ± 1.07	47.90 ± 1.17	27.47 ± 0.95	–
Ni/AC C	23.59 ± 0.76	–	4.89 ± 1.95	71.82 ± 0.98
Ni/AC M	20.98 ± 3.61	–	11.14 ± 0.30	67.89 ± 3.51
Ni/AC S	21.86 ± 2.88	–	4.32 ± 0.77	73.82 ± 3.65

* Values are reported as mean ± standard deviation (n = 3).

clearly shows the most preserved pore structure. This finding is in agreement with BET findings, whereby the AC supported catalyst showed higher BET surface area due to its porous structures (see Table 3). Ni/AC M demonstrates more finely dispersed surface particles, and Ni/AC S displays localized pore blockage. These features align with the EDX maps, Ni/AC C displays uniform Ni coverage, Ni/AC M shows finely dispersed Ni, and Ni/AC S presents uneven distribution and localized agglomeration. Oxygen, O, detected in Ni/AC catalyst elemental mapping can be attributed to 1) NiO traces species as confirmed by XRD and H_2-TPR and/or 2) O in surface groups that resulted from the acidification of AC with H_2SO_4.

3.2. Hydrocarbon liquid product characterization

The hydrocarbon yield of blank reaction (no catalyst) and catalyzed reactions are presented in Fig. 1(c). Blank reaction is observed to have

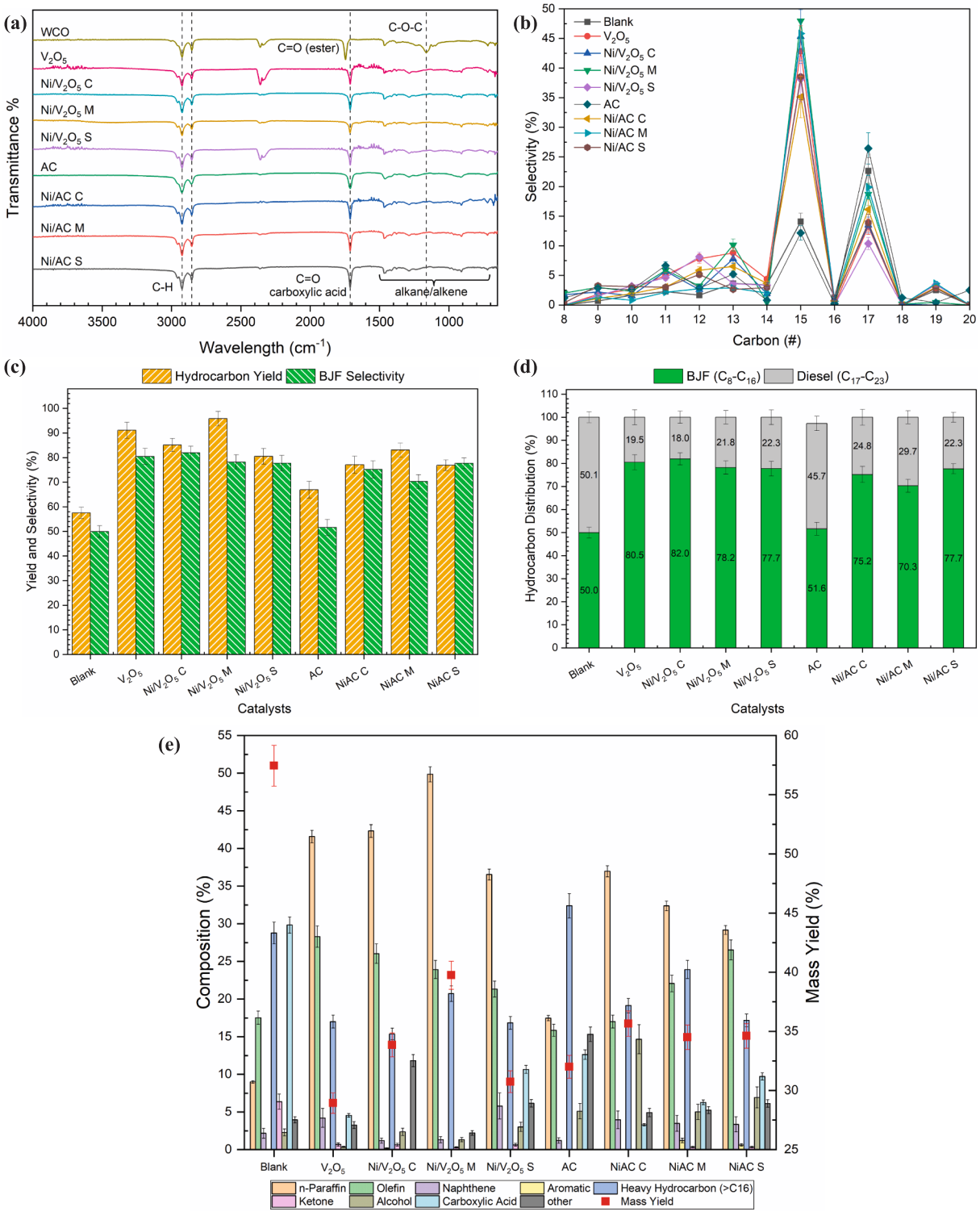


Fig. 1. (a) FTIR spectra showing functional groups composition, (b) carbon number distribution, (c) hydrocarbon liquid yield, (d) hydrocarbon distribution based on carbon number and (e) product distribution and mass yield liquid product from DO of WCO at 400 ℃ over 5 wt% loading of various catalysts for 2 h.

produced the lowest yield of hydrocarbons (58%), while catalyzed reactions significantly yielded higher contents of hydrocarbon (67–96%) ranked in the order Ni/V_2O_5 M > V_2O_5 > Ni/V_2O_5 C > Ni/AC M > Ni/V_2O_5 S > Ni/AC C > Ni/AC S > AC. Similarly on the same figure, for jet range hydrocarbon percentage (C_8-C_{16}), catalytic reactions showed higher selectivity (52–82%) than blank reaction (50%). Notably, the high yield of BJF highlights strong selectivity for simultaneous DO-cracking pathways, supported by the carbon number distribution depicted in Fig. 1(b), where catalyzed reactions showed highest concentrations of C_{15} likely resulting from 1) decarboxylation/decarbonylation ($DeCO_x$) of palmitic acid, 2) hydrodeoxygenation (via in-situ H_2)/$DeCO_x$ and cracking of oleic acid. Note, V_2O_5 and Ni/V_2O_5 catalyst series gave rise to a noticeable higher BJF fractions (Fig. 1(d)) owing to the presence of rich medium + strong acidic sites which was corroborated with the fact that increase in only strong acidic sites had detrimental effects on BJF selectivity as can be observed with the Ni/AC catalyst series. In general Ni/V_2O_5 catalyst series outperformed their Ni/AC counterparts showing high DO activity likely due to the presence of rich oxygen vacancies (O_v) from the vanadium oxide support. The Ni-O-V interfacial sites can be analogous to Ni-O_v that are known to act as adsorption sites, with string affinity for carboxyl (–COOH) and carbonyl (C═O) groups, boosting DO activity (Liu et al., 2023; J. Wang et al., 2023). This high concentration of O_v enhances the uniform dispersion of Ni across the support (Zhou et al., 2025). It can be concluded that the smaller crystallite size of Ni/V_2O_5 M (see section 3.1), led to a higher density of lattice imperfections which in turn promoted the formation of more O_v sites as corroborated by XRD and H_2-TPR results and therefore resulted in enhanced DO activity. Indeed, AC as a standalone catalyst showed limited DO activity yet resulted in a high fraction of oxygenated intermediates mainly fatty aldehydes and carboxylic acids. However, Ni metal doping on AC as a support was observed to yield superior DO activity, highlighting the essential role of Ni in promoting C-—O bond cleavage. Nonetheless, Ni/AC catalyst series lack oxygen lattice, even though trace amounts of NiO are present, there is effectively limited O_v that exist. Therefore, Ni/AC catalyst relies mainly on its metallic Ni sites and high acidity to cleave C-—O but also risks faster deactivation, in the absence of H_2, by coke deposition leading to severe blockage of active Ni sites.

Chemical composition is paramount to the quality of the fuel, it can also be utilized to determine the effectiveness of catalyzed reactions in terms of DO activity or the extent of oxygen-containing compounds removal, as shown in the product distribution results (Fig. 1(e)). Blank experiment yielded significantly high carboxylic acid indicating weak DO of intermediate free fatty acids, it can also be deduced that C–—C cleavage did not occur a great deal as evident from the presence of high heavy hydrocarbons (>C_{16}). Catalyzed reactions predominantly yielded *n*-paraffins (alkanes) and olefins (alkenes) jet range hydrocarbons in the range C_8-C_{16}. Ni/V_2O_5 M, Ni/V_2O_5 C and V_2O_5 were able to produce highest *n*-paraffin content of 50%, 42% and 42% respectively, while blank yielded lowest *n*-paraffin. It is well known that during non-catalyzed thermal cracking of WCO, more olefins than paraffins are typically produced, mainly due to the breakdown of unsaturated fatty

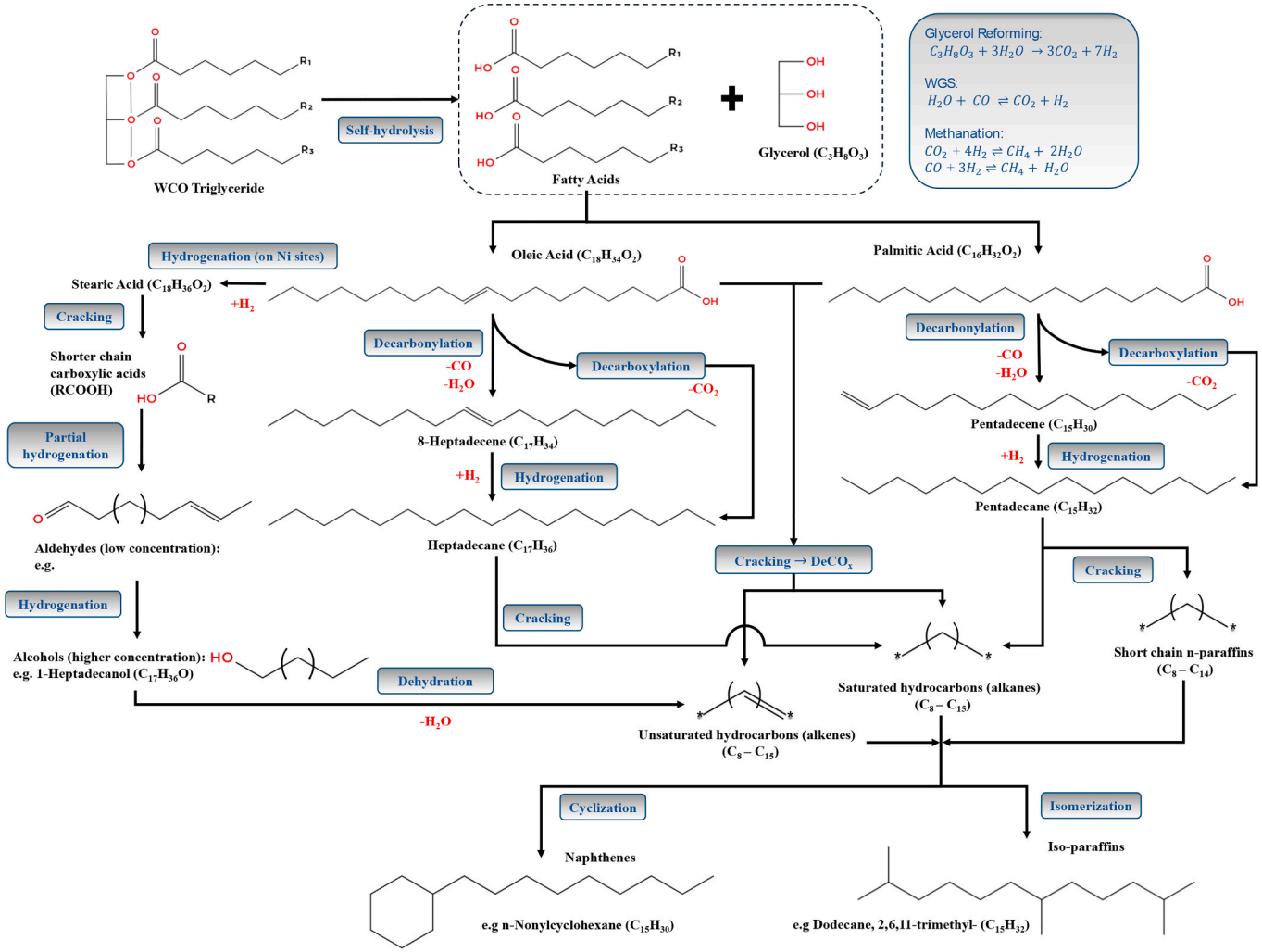


Fig. 2. Proposed reaction mechanism for WCO catalytic deoxygenation over Ni/V_2O_5 catalyst.

acids which are abundant in WCO (Hebish et al., 2024). Notably, Ni/V_2O_5 catalyst series yielded lower olefins compared to non-modified V_2O_5 while favouring more paraffin content, suggesting strong hydrogenation Ni metal sites were able to saturate C═C. Studies show that strongly acidic catalysts tend to give alkenes/cycloalkenes as primary (Hu et al., 2019), though here V_2O_5 still yielded substantial alkanes, likely due to secondary hydrogenation as acid sites can isomerize/crack intermediates, but metal sites and/or in-situ hydrogen shift alkenes to alkanes (del Campo et al., 2021). It is worth noting that olefins are prone to oxidative instability and degradation during storage, as olefins readily react with oxygen, forming gums and sediments (Why et al., 2022). High olefin content will cause oxidation to the fuel therefore it is not favourable. In general Ni/AC catalyst series produced good outcomes, however owing to their high acidic nature, coke formation is highly promoted which accelerates catalyst deactivation as evident from literature (Li et al., 2017). In terms of total by-products, ketone, alcohol, carboxylic acid and others, blank yielded the highest percentage of 43%, while Ni/V_2O_5 M yielded the lowest percentage of only 4%.

Based on the product composition obtained from GC–MS result (liquid-phase) and GC-TCD (gas-phase), the proposed reaction pathway can be outlined as shown in Fig. 2. WCO feedstock was mainly comprised of long chain fatty acids including 34% palmitic acid (C16:0) and 59% oleic acid (C18:1) (see Table 1). Initially, triglycerides in WCO undergo self-hydrolysis via hydrogen abstraction to form glycerol and free fatty acids (FFAs), mainly palmitic acid and oleic acid. The FFAs adsorbed on Ni/V_2O_5 surface underwent DO through decarboxylation (DCX), decarbonylation (DCO), hydrogenation and cracking pathways. High concentration of C_{2n+1} hydrocarbons detected in the product was consistent with the predominance of odd carbon number hydrocarbons typically observed during H_2-free DO, indicating that DCO, DCX and cracking routes dominate over direct HDO (Zulkepli et al., 2018). For palmitic acid, DCO proceeds yielding pentadecene ($C_{15}H_{30}$) as an intermediate, after which the C═C bond is subsequently hydrogenated on Ni sites to produce pentadecane ($C_{15}H_{32}$). Alternatively, palmitic acid can undergo decarboxylation, directly producing pentadecane without olefinic intermediates. Similarly for oleic acid, the DCO pathway yields 8-Heptadecene ($C_{17}H_{34}$) which is further hydrogenated to heptadecane ($C_{17}H_{36}$) whereas the DCX route directly produces heptadecane. In addition to direct DO, secondary cracking reactions via C–—C cleavage of fatty acids and long-chain hydrocarbons occur, followed by DO, leading to the formation of a mixture of shorter-chain alkanes and alkenes. Another minor but detectable pathway involves the hydrogenation of oleic acid to stearic acid (C18:0), followed by C–—C bond scission and partial hydrogenation to oxygenated intermediates. This route is supported by the presence of trace amounts of E-15-Heptadecenal ($C_{17}H_{32}O$). These aldehydes can be further hydrogenated to alcohols such as 1-Heptadecanol ($C_{17}H_{36}O$), which may undergo dehydration over acidic sites to generate short-chain olefins (Nugraha et al., 2024). At the elevated reaction temperatures, simultaneous C–—C bond scission via β-scission and mild thermal cracking occurs, particularly in the presence of acidic V_2O_5 support sites, which promotes formation of shorter chain hydrocarbons (such as 1-octene, octane, 1-nonene, nonane, 1-decene, decane, 1-undecene and undecane) and shorter chain FFAs, which are then deoxygenated. Note, secondary reactions such as water–gas shift (WGS), glycerol reforming and methanation generate in-situ H_2, enabling hydrogenation of unsaturated fatty acids into saturated intermediates and olefins into saturated hydrocarbons (Asikin-Mijan et al., 2022). This in-situ H_2 phenomenon explains the relatively high alkane-to-alkene ratios observed under H_2-free DO conditions (Nugraha et al., 2024). Furthermore, isomerization, and limited cyclization reactions may further transform linear hydrocarbons into branched and cyclic species. In line with this, our results show small amounts of cyclic compounds (naphthenes), such as *n*-Nonylcyclohexane ($C_{15}H_{30}$) reaching up to 6%, consistent with previous studies (Zulkepli et al., 2018). In summary, these results highlight the effectiveness of Ni/V_2O_5 in promoting the efficient DO of –COOH fragments through the $DeCO_x$ route while minimizing excessive cracking which would otherwise result in gaseous light hydrocarbons out of BJF range. The gaseous products were evaluated by GC-TCD and depicted in Fig. S6. The gas distribution consisted primarily of CO (51.2%), followed by CO_2 (21.3%), H_2 (20.3%), and a minor amount of CH_4 (7.1%). CO and CO_2 detected are directly associated with deoxygenation via DCO and DCX pathways, respectively. Indeed, the higher content of CO compared to CO_2 indicates the oxygen removal occurred mainly via DCO pathway. At the same time, CH_4 was detected which can be attributed to secondary cracking and methanation reactions (Asikin-Mijan et al., 2022). Interestingly, although a fair amount of H_2 was detected in the gas phase, the reaction is not favoured to have proceeded primarily through the HDO pathway, this is supported by the dominance of odd-numbered hydrocarbons in the liquid product. The detected H_2 was likely generated in-situ via dehydrogenation by WGS reaction and may have contributed to the hydrogenation of intermediate unsaturated hydrocarbons, leading to the high selectivity towards saturated alkanes observed in the liquid product.

Fig. 1(a) illustrates the use of spectroscopic Fourier Transform Infrared Spectroscopy (FTIR), for the qualitative characterization of chemical functional groups present in waste cooking oil (WCO) and its deoxygenated products. The FTIR spectrum of WCO exhibited distinct absorption bands at 3007 cm^{-1}, 2924 cm^{-1}, 2855 cm^{-1}, and 1465 cm^{-1}, along with an additional peak at 1377 cm^{-1}, corresponding to the stretching and bending vibrations of aliphatic C–—H bonds (Hafriz et al., 2022). Additionally, absorption peaks at 1746 cm^{-1} and 1165 cm^{-1} were attributed to the carbonyl (C═O) group of glycerides and the C–—O stretching from ester bonds in WCO, respectively (Khalit et al., 2021). The deoxygenation process was evidenced by a slight shift of the carbonyl (C═O) absorption band from 1745 cm^{-1} in WCO to 1713 cm^{-1} in the deoxygenated product, along with a notable decrease in the intensity of the C–—O stretching band (at 1163 cm^{-1} in WCO). This spectral transformation indicates the progression of the DO reaction. The decrease in peak intensities correlated well with the reduction in oxygen content observed in the final product (Aliana-Nasharuddin et al., 2020). The FTIR analysis of the deoxygenated liquid products revealed the presence of alkane and alkene compounds, as indicated by C–—H stretching vibrations at 2856 cm^{-1} and 2924 cm^{-1}. Furthermore, the emergence of bending vibrations at 1381 cm^{-1} and 1462 cm^{-1} for alkane –C–—H and at 721 cm^{-1} and 723 cm^{-1} for alkene –CH═CH– confirms the formation of hydrocarbon products following the DO process, with more prominent bands for alkane, indicating higher composition. Furthermore, the absence of a broad, strong peak around 3400 cm^{-1} for O–—H stretching and another distinct peak of H-O–—H bending around 1640 cm^{-1} indicated that no residual water is present across all liquid products. This observation confirms that the sample are essentially moisture-free as presence of water in the fuel has negative effects on engines and fuel systems and therefore hazardous on flight safety. Such effects are well documented in the literature which includes, accelerated corrosion, poorer fuel lubricative properties, microbial growth, degrades additives and antioxidants and restrict or potentially block fuel flow in fuel systems due to icing (Chen et al., 2022). Overall, the microwave synthesized Ni/V_2O_5 catalyst was the most efficient catalyst in converting the WCO, producing high content of hydrocarbons and selective towards BJF fraction, with minimal unwanted by-products.

3.3. Performance comparison of non-noble metal catalyst for WCO deoxygenation with present study

The deoxygenation performance obtained in this study is compared with similar Ni-based catalysts/WCO feedstocks systems from previous literature (Table 5). In this study, using a microwave-assisted Ni/V_2O_5 catalyst, a 96% hydrocarbon yield with 78% BJF selectivity was obtained from the DO of WCO at 400 ℃ over 5 wt% for 2 h. Hafriz et al., 2022 utilized a low-cost NiO dolomite-based catalyst on WCO, however

Table 5
Comparison between existing literature and the current study on BJF production from WCO under Ni-based catalysts.

No	Catalyst	Feed	Reaction Conditions	Catalyst Stability	Hydrocarbon Yield (%)	BJF Selectivity (%)	Ref.
1	Ni/V_2O_5	WCO	T = 400 ℃, t = 2h, catalyst loading = 5 wt%, metal% = 20 wt% Ni, N_2 gas at 20 mL/min	E_d[a] = 48.3 kJ/mol, stable > 8h TOS[b]	96	78	Present study
2	NiO-Dolomite	WCO	T = 400 ℃, t = 0.875 h, catalyst loading = 3.25 wt%, metal% = 5 wt% Ni N_2 gas at 175 mL/min	NR[c]	61	72	(Hafriz et al., 2022)
3	5Ni5Co/SBA-15-SH	WCO	T = 350 ℃, t = 2h, catalyst loading = 5 wt%, metal% = 5 wt% Ni, 5 wt% Co, N_2 gas at 20 mL/min	stable up to 3rd cycle	80	92	(Razak et al., 2024)
4[d]	Ni-Mo/Al_2O_3-SiO_2	WCO	T = 350 ℃, WHSV = 1.1 h^{-1}, metal% = 3 wt% Ni, 10 wt % Mo, H_2 gas at 60 bar	stable > 100 h TOS	87	75	(Racha et al., 2024)
5[d]	Ni-Mo/Al_2O_3-SiO_2	WCO	T = 420 ℃, LHSV = 1h^{-1}, H_2/oil = 2200 NL/L at 100 bar	stable > 220 h TOS	97	56	(Verma et al., 2023)
6	Ni/palygorskite	WCO	T = 310 ℃, t = 9h, metal% = 30 wt% Ni, catalyst loading = 1 wt%, H_2 gas at 40 bar	NR	82	NR	(Lycourghiotis et al., 2019)
7	Ni/*Meso*-Y	WCO	T = 400 ℃, t = 8h, catalyst loading = 5 wt%, metal% = 10 wt%, H_2 gas at 30 bar	NR	NR	41	(Li et al., 2015)
8[d]	Ni_2P/AC	WCO	T = 300 ℃, 1.2 wt% in *n*-heptane, GHSV = 139.8 h^{-1}, H_2 at 1 bar	NR	77	NR	(Pham et al., 2019)
9	Ni/AC	WCO	T = 350 ℃, t = 3h, catalyst loading = 5 wt%, metal% = 20 wt% Ni, N_2 gas at 10 mL/min	stable up to 4th cycle	90	66	(Khalit et al., 2021)
10	$Ni/CeZrO_2$	Oleic Acid	T = 300 ℃, t = 3h, catalyst loading = 3 wt%, metal% = 20 wt%, 20% H_2/N_2 at 1 bar	NR	98	20	(Jeon et al., 2019)
11[d]	Ni/Mo ZrO_2	CPO[e]	T = 400-500℃, WHSV = 5–25 h^{-1}, H_2 at 60 mL/min	E_d = 54.3 kJ/mol	NR	NR	(Hasanudin et al., 2019)

[a] E_d = Activation energy of deactivation (see section 3.4).
[b] catalyst retains > 70% of its catalytic activity after 8 h TOS.
[c] NR = Not reported.
[d] Fixed bed reactor.
[e] CPO = crude palm oil.

the catalyst exhibited lower hydrocarbon yield and selectivity due to its limited acidity and thermal stability as structure collapses of dolomite catalyst occurs at high temperatures (>400 ℃). Several studies investigated the HDO of WCO over Ni-Mo/alumina-silica catalyst. For instance, Verma et al., 2023 reported 97% yield and 56% jet-range selectivity. In a similar catalyst system, Racha et al., 2024 attained an 87% yield and 75% selectivity towards jet-range, attributed to the enhanced acidity and good metal dispersion on the mixed oxide supports. Despite this, the process requires high hydrogen pressure which increases costs and system complexity. Razak et al., 2024 reported the DO of WCO using a 5Ni5Co/SBA-15-SH catalyst producing 80% of yield with 92% of bio-jet selectivity, which was credited to the high surface area and enhanced acidity of the mesoporous SBA-15 support. Nonetheless, the catalyst preparation involves multiple synthesis and functionalization steps which may increase production costs and limit industrial scale-up. Therefore, the Ni/V_2O_5 catalyst developed in this study demonstrates strong potential for commercial production of BJF due to its use of inexpensive materials, simple preparation method, good catalyst stability and superior catalytic activity.

3.4. Kinetic deactivation study on DO of WCO over Ni/V_2O_5 microwave-synthesised catalyst

The catalyst stability of Ni/V_2O_5 M was evaluated at 380—420 ℃. The Arrhenius plot of $\ln k_d$ against 1/T was constructed based on Eq.6 producing a linear relationship from which the activation energy of deactivation (E_d) was determined (see Fig. 3 (d)). The strong linearity obtained (R^2 = 0.94) indicates that the deactivation rate constant follows Arrhenius behaviour and that the reaction kinetics are well described by Arrhenius equation. From the graph, the E_d of Ni/V_2O_5 M in the DO of WCO was calculated to be 48.3 kJ/mol. This value is lower than that reported by Mohmmed et al. of 62.4 kJ/mol for the DO of oleic acid over coated Pd/AC (Mohammed et al., 2022) and by Hasanudin et al. of 54.3 kJ/mol for the HDO of crude palm oil over Ni/Mo ZrO_2(Hasanudin et al., 2019). The lower E_d value of this study indicates that the catalyst decay of Ni/V_2O_5 is much less temperature sensitive and therefore the catalyst is more stable and robust. Note, Fig. 3b-c shows the weight loss curve in the TGA/O_2 analysis and RAMAN spectra of spent catalyst after kinetic study, respectively. The weight loss observed in the 200–400 ℃ region is attributed to the combustion of soft amorphous coke which is known to strongly poison catalyst active sites and thus cause rapid loss of catalyst activity (Li et al., 2017). In contrast, weight loss occurring at temperatures above 400 ℃ is assumed to resemble hard coke of graphitic structure, although these deposits are more difficult to remove during regeneration, they are generally much less immediately deactivating (Shakor et al., 2024). Hard coke predominated the carbon deposits on Ni/V_2O_5 as evident from the weight drop after 400℃. The RAMAN spectrum provided further complementary evidence, which displayed a pronounced G band (~1596 cm^{-1}) and an extremely weak D band (~1350 cm^{-1}). Obviously, the low D/G intensity ratio points to the presence of more ordered graphitic carbon. These findings indicate that catalyst deactivation was mainly caused by carbon deposition of catalysts surface, which likely blocked active sites and reduced catalytic activity.

4. Conclusions

In this study, the catalytic deoxygenation of WCO to jet-like hydrocarbons under H_2-free conditions was investigated and compared over a series of transition metal-based catalysts, including a novel Ni/V_2O_5 catalyst, synthesised via wet-impregnation and subjected to different treatments. The synthesized catalysts by microwave- and sonication-assistance demonstrated improved physico-chemical properties which contributed to their high reaction performance. These enhanced methods reduced synthesis time and energy consumption while delivering comparable or superior catalytic performance. Ni/V_2O_5 M exhibited the best performance with a hydrocarbon yield of 96% and a jet-selectivity of 78% consisting mainly of desirable *n*-paraffins. The synergistic effect between Ni and V_2O_5 lead to formation of rich oxygen vacancies that enhanced the $DeCO_x$ activity and produced favourable

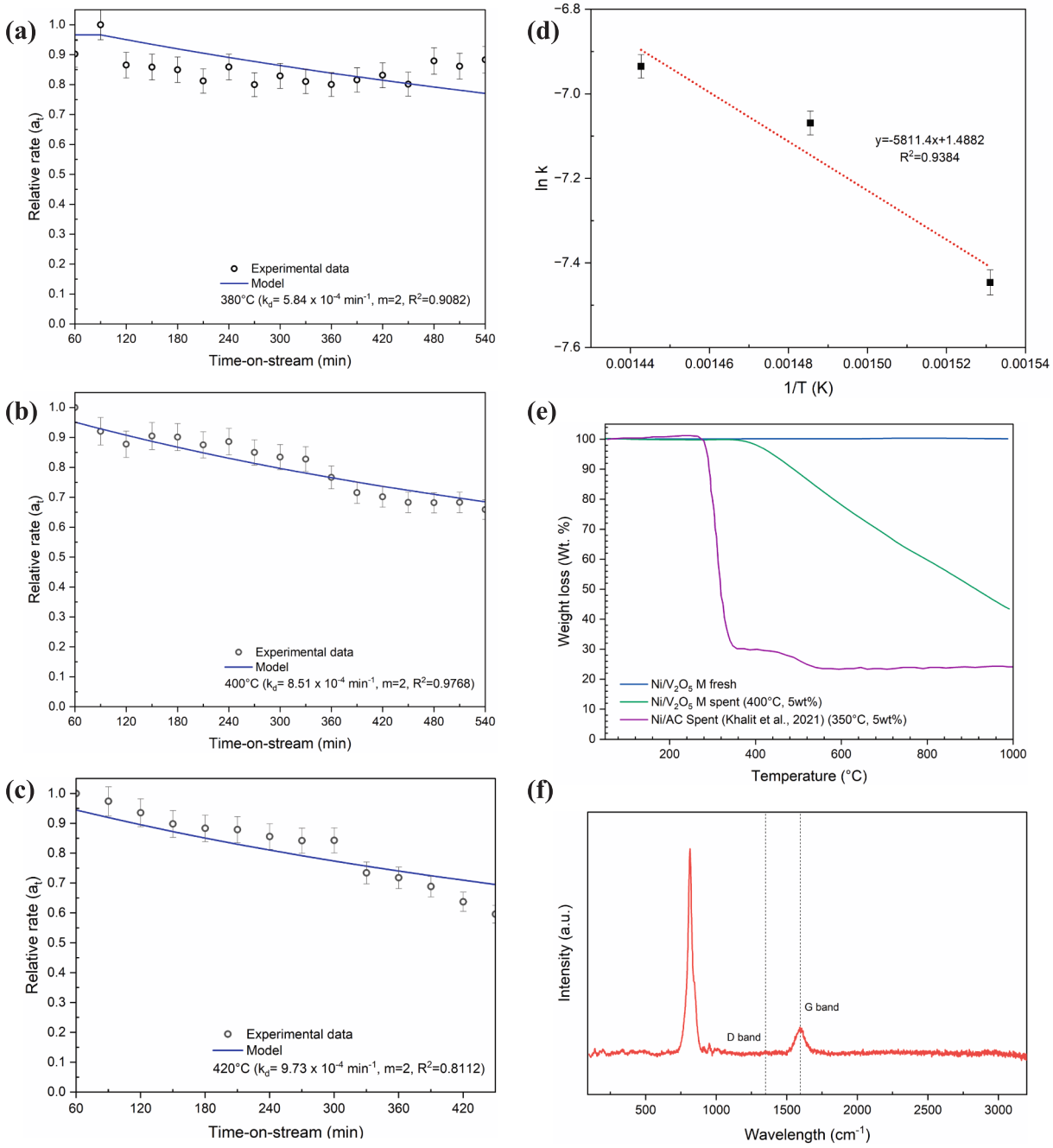


Fig. 3. Plot of relative rate (at) against time-on-stream (min) and the second-order power-law deactivation model fitting for the DO of WCO over Ni/V2O5 M at different temperatures (a) 380 °C, (b) 400 °C and (c) 420 °C, (d) Arrhenius Plot for the DO of WCO over Ni/V_2O_5 M and (e) TGA analysis of fresh and spent catalyst, (f) RAMAN spectrum of spent Ni/V_2O_5 M catalyst.

selectivity. Kinetic results established that deactivation of Ni/V_2O_5 catalyst is resilient to temperature changes and therefore possess high stability. Therefore, given its low-cost and high activity, Ni/V_2O_5 catalyst is promising for the industrial scale-up and the maturity of the DO of WCO in the absence of H_2. In future studies, the enhanced properties achievable through microwave and sonication assistance should be consistently exploited and even extended to existing catalyst systems to maximize catalytic performance and increase efficiency.

CRediT authorship contribution statement

Anas Ahmed Hebish: Writing – review & editing, Writing – original draft, Methodology, Investigation, Formal analysis. **Nurin Wahidah Mohd Zulkifli:** Writing – review & editing, Supervision, Resources, Conceptualization. **Nurhidayatullaili Muhd Julkapli:** Writing – review & editing, Supervision, Conceptualization. **Muhamad Fazly Abdul Patah:** Writing – review & editing, Supervision, Conceptualization. **N. Asikin-Mijan:** Writing – review & editing, Validation, Supervision, Resources, Methodology, Conceptualization. **Syawal Mohd Yusof:** Resources. **Salma Samidin:** Resources, Investigation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.biortech.2026.134969.

Data availability

Data will be made available on request.

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